Quantum Electrodynamic Treatment of Harmonic Generation in Intense Optical Fields

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A theoretical description of high order harmonic generation has been developed that draws upon QED scattering theory and relationships developed previously for multiphoton processes. The framework presented here offers a new perspective on the harmonic generation process and reproduces the salient characteristics of experimental spectra. A direct connection is drawn between harmonic generation and above threshold ionization, and calculated spectra suggest that transitions between (stationary) quantized field Volkov states represent the dominant contribution to the spectrum. [S0031-9007(98)06891-4]

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A decade ago, McPherson and co-workers [1] demonstrated that atoms subjected to an intense optical field ($I \gtrsim 10^{13}$ W cm^{-2}) generate high order odd harmonics of that field. Several theoretical approaches to describing this phenomenon by nonperturbative methods have been pursued [2–8], all of which are semiclassical in the sense that the optical field is treated classically. Harmonic spectra have been calculated by Krause, Schafer, and Kulander [3], and Lewenstein *et al.* [6] by solving Schrödinger's equation, but theoretical and experimental results have generally been interpreted in terms of the semiclassical two step model [2,4] in which the bound electron tunnels into the continuum under the influence of an intense optical field, resulting in the formation of a continuum wave packet. Radiation is produced when a portion of the wave packet subsequently returns to the ion core.

An alternative theoretical perspective for viewing the processes underlying high order harmonic generation (HG) is presented here. Drawing from the nonperturbative QED scattering theory formalism recently developed for multiphoton ionization (MPI) and above threshold ionization (ATI) [9], the transition matrix describing the interaction of an atom with a quantized optical field through quantum Volkov states [10] is derived and its implications are discussed. In this view, the electron-photon interaction is dominant and, although energy and momentum conservation require that the free electron ultimately return to the ground state of the atom, transitions between Volkov states represent the primary contribution to HG spectra.

Keldysh-Faisal-Reiss (KFR) theory [11] offers a nonperturbative framework for MPI that incorporates classical Volkov states which are solutions of the time-dependent Schrödinger equation for a free electron in an electromagnetic field. Recently, Guo, Åberg, and Crasemann [9] and Guo and Drake [12] developed a QED description of MPI involving *quantized field* ("quantum") Volkov states and stressed the advantages of this approach, one of which is that the electron and the photon field are treated as an isolated system; thus, the energy and momentum of the system are well defined. The transition matrix element T_{fi} describing the atom-optical field interaction in which an electron, initially in the bound state Φ_i, is excited to the intermediate quantum Volkov state(s) Ψ_μ and, ultimately, to the final state Φ_f (that is decoupled from the field or "field free") is given by the relation [9]

$$T_{fi} = \sum_\mu \langle \Phi_f; m \,|\, \Psi_\mu \rangle \langle \Psi_\mu | V | \Phi_i; n \rangle , \qquad (1)$$

where m and n represent the number of photons in the pump laser field.

Although Eq. (1) provides an accurate description of MPI, it is not suitable for exploring HG because secondary photon modes are not included and, hence, no mechanism for generating harmonics is provided. In order to account for the interaction of the fundamental optical field with secondary (harmonic) photon modes, the Volkov state is modified to include a perturbative term [13]. The low intensities of the generated harmonics relative to that for the fundamental, as well as the stationary nature of the quantum Volkov states, allow one to invoke time-independent perturbation theory, from which the perturbed Volkov wave function is found to be

$$|\Psi'_\mu\rangle = \sum_{P',n'} \frac{\langle \vec{P}', n'; 1 | V' | \vec{P}, n; 0 \rangle}{E' - E} \, |\vec{P}', n'; 1\rangle , \qquad (2)$$

where the initial and intermediate Volkov states are given, respectively, by the expressions [12]

$$|\vec{P}, n; 0\rangle = V_e^{-1/2} \exp\{i[\vec{P} + z\vec{k}] \cdot \vec{r}\} \exp\{-iN_a \vec{k} \cdot \vec{r}\} \times \exp\{-F\} \, |n\rangle \, |0\rangle , \qquad (3)$$

$$|\vec{P}', n'; 1\rangle = V_e^{-1/2} \exp\{i[\vec{P}' + z\vec{k}] \cdot \vec{r}\} \exp\{-iN_a \vec{k} \cdot \vec{r}\} \times \exp\{-F'\} \, |n'\rangle \, |1\rangle , \qquad (4)$$

with energy eigenvalues of $E = (\vec{P}^2/2m) + n\omega + z\omega$, and $E = (\vec{P}'^2/2m) + n'\omega + z\omega + \omega'$, where $\hbar \equiv c \equiv 1$, and ω and ω' are the fundamental (pump) and harmonic frequencies, respectively. In Eqs. (2)–(4), $\vec{P}$ is the momentum vector of the electron, N_a is the number operator for the pump field, V_e is the normalization volume of the electron, z is the ponderomotive energy

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($U_p = e^2E^2/4m\omega^2$) normalized to the fundamental photon energy, and the coupling between the electron and laser field is represented by the operator $\exp(-F)$, where

$$F = \frac{eg}{m\omega}(\vec{P}\cdot\hat{\varepsilon}\hat{a}e^{i\Theta} - \vec{P}\cdot\hat{\varepsilon}\hat{a}^{\dagger}e^{-i\Theta}) + \frac{e^2g^2\cos\xi}{4m\omega}(e^{-i2\Theta}\hat{a}^2 - e^{i2\Theta}\hat{a}^{\dagger 2}), \quad (5)$$

g is the normalization factor arising from quantization of the field, $\hat{\varepsilon}$ is the polarization vector, and ξ expresses the degree of polarization of the optical field. The initial phase of the laser field, Θ, is defined in relation to the vector potential:

$$\vec{A}(\vec{k}\cdot\vec{r}) = g[\hat{\varepsilon}\exp(i\vec{k}\cdot\vec{r})e^{i\Theta}\hat{a} + \hat{\varepsilon}^*\exp(-i\vec{k}\cdot\vec{r})e^{-i\Theta}\hat{a}^{\dagger}], \quad (6)$$

and, in the calculations to follow, the driving optical field is assumed to be linearly polarized ($\xi = 0$). Notice that, in Eqs. (2)–(4), the intermediate Volkov state wave function incorporates new values of electron momentum, photon number, and phase (Θ'). Also, one photon is now generated in the secondary mode (harmonics) of the field.

Consequently, the transition matrix element describing the entire HG process can be expressed as

$$T_{fi} = \sum_{\vec{P}',n',\vec{P},n}\langle\Phi_f,m;1|\vec{P}',n';1\rangle\frac{\langle\vec{P}',n';1|V'|\vec{P},n;0\rangle}{\Delta E} \times\langle\vec{P},n;0|V|\Phi_i,l;0\rangle \quad (7)$$

in which the two interaction Hamiltonians are given by the relations

$$V = -\frac{e}{m}(-i\vec{\nabla})\cdot\vec{A}(\vec{k}\cdot\vec{r}) + \frac{e^2A^2(\vec{k}\cdot\vec{r})}{2m}, \quad (8)$$

and

$$V' = -\frac{e}{m}(-i\vec{\nabla})\cdot\vec{A}'(\vec{k}'\cdot\vec{r}) + \frac{e^2}{m}\vec{A}(\vec{k}\cdot\vec{r})\cdot\vec{A}'(\vec{k}'\cdot\vec{r}). \quad (9)$$

In general, the final (field-free) state Φ_f can be that of a free electron or a bound state of an atom. As discussed by Becker *et al.* [8], the KFR approach to *ionization* includes approximating the final scattering state of a photoelectron by a plane wave. In contrast, the conservation of energy and momentum in *harmonic generation* requires that Φ_f be the initial (bound) state Φ_i. Equating the initial and final states of the HG transition matrix,

$$E(|\Phi_f\rangle) + m\omega + \omega' = E(|\Phi_i\rangle) + l\omega, \quad (10)$$

yields

$$\omega' = (l - m)\omega \equiv q\omega, \quad (11)$$

where q is the harmonic order.

It is the middle term of Eq. (7) that is instrumental in describing and calculating the HG spectrum. This matrix element, denoted T^V, represents transitions between quantum Volkov states and can be expanded as

$$T^V = \langle\vec{P}',n';1|V'|\vec{P},n;0\rangle = \frac{e^2g'}{m}\frac{(2\pi)^3}{V_e}\delta(\vec{P}' + n'\vec{k} + \vec{k}',\vec{P} + n\vec{k}) \times\langle n'|\exp(F')[\vec{P} - eg\hat{\varepsilon}(a + a^{\dagger})]\cdot\hat{\varepsilon}'\exp(-F)|n\rangle, \quad (12)$$

where the δ function enforces momentum conservation during the Volkov → Volkov transition. When combined with momentum conservation, the resonance condition ($\Delta E \to 0$) determines the constraints on harmonic generation: $\omega' = (n - n')\omega \equiv q\omega$, and $\vec{k}' = (n - n')\vec{k} \equiv q\vec{k}$.

Summing over all intermediate Volkov states $|\vec{P}',n'\rangle$ requires integrating over $\vec{P}'$, which results in the following expression [14]:

$$\frac{V_e}{(2\pi)^3}\int d\vec{P}'^3\langle\vec{P}',n';1|V'|\vec{P},n;0\rangle = \frac{eg'\omega}{2\Lambda}\left\{-\zeta\mathfrak{T}_{-q}\left(x,y,-\frac{\pi}{4}\right) + 2ze^{-i\frac{\Delta\Theta+\pi}{2}}\mathfrak{T}_{-q+1}\left(x,y,-\frac{\pi}{4}\right) + 2ze^{i\frac{\Delta\Theta+\pi}{2}}\mathfrak{T}_{-q-1}\left(x,y,-\frac{\pi}{4}\right)\right\}, \quad (13)$$

where $\Delta\Theta = \Theta' - \Theta$ is the shift in the phase of the pump optical field that occurs during transitions between the Volkov states $|\vec{P},n\rangle$ and $|\vec{P}',n'\rangle$, $\mathfrak{T}_j(x,y,\phi)$ is the generalized Bessel function, $\zeta = (2e\Lambda/m\omega)\vec{P}\cdot\hat{\varepsilon}$ where 2Λ is the amplitude of the classical vector potential in the limit of large photon number, $x = 2\zeta\sin(\Delta\Theta/2)$ and $y = -z\sin(\Delta\Theta)$.

Equation (13) is the expression for the Volkov → Volkov state transition matrix element, given $\vec{P}$. Because of the energy conservation relations discussed earlier, $|\vec{P}|$ takes on only the discrete values:

$$|\vec{P}| = P_j = \sqrt{2m\omega(j - z - E_b/\omega)}, \quad (14)$$

where E_b is the ionization energy of the atom or molecule of interest and j is an integer representing the number of photons absorbed in the MPI process. Therefore, $\zeta_j = (2e\Lambda/m\omega)P_j\cos\varphi$, where φ is the angle between $\vec{P}$ and $\hat{\varepsilon}$. Several characteristic features of HG spectra are best illustrated by first calculating

$$I_q^V = \int_0^{2\pi} d(\Delta\Theta) \times\left|\frac{V_e}{(2\pi)^3}\int d\vec{P}'^3\langle\vec{P}',n' = n - q;1|V'|\vec{P},n;0\rangle\right|^2 \quad (15)$$

for several values of j. Results that are representative of those obtained over a range in experimental conditions are shown in Fig. 1 for Ne($E_b \simeq 21.6$ eV) and a laser wavelength and peak intensity of 0.8 μm and 1.3×10^{15} W cm^{-2}, respectively. For these parameters,

$\hbar\omega \simeq 1.55$ eV, $U_p = 78.8$ eV, and ionizing the atom requires a minimum of 66 photons. Choosing $\vec{P}$ to lie along $\hat{\varepsilon}$, Figs. 1(a)–1(c) show the harmonic spectra generated for $j = 66$, 67, and 70, which correspond to photoelectron kinetic energies of 1.24, 2.24, and 5.24 eV, respectively. Figure 1(d) shows the effect on the $j_{\min} = 66$ spectrum [Fig. 1(a)] of reducing the pump intensity by a factor of 2. Despite the fact that only the contribution of the quantum Volkov → Volkov transition matrix element to the overall HG spectrum is illustrated in Fig. 1, all of the spectra reproduce the plateau and cutoff that are observed experimentally. Cutoff arises from a property of the generalized Bessel function $\mathfrak{T}_q(x, y, \phi)$ which, when x and y are fixed, drops off sharply above $q_{\text{cutoff}} \simeq |x| + |2y|$. One concludes that each ATI order $j - j_{\min} + 1$ (which corresponds to a specific photoelectron energy) *has an associated HG spectrum* that is characterized by a cutoff defined by $q_{\text{cutoff}} \cong 2z + 2\sqrt{z(j - z - E_b/\omega)} \simeq 2z$ for $U_p \gg E_b$ and low values of j (i.e., $j \simeq j_{\min}$).

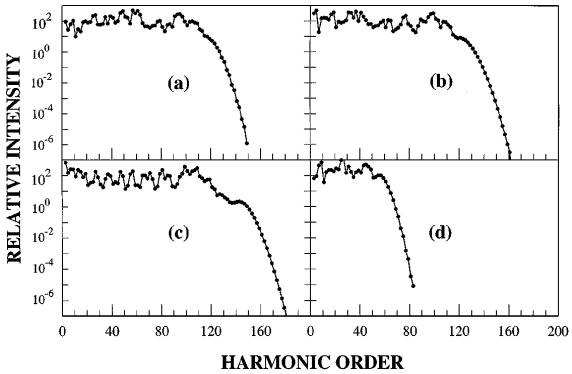


FIG. 1. Harmonic spectra calculated from Eq. (15) for a laser wavelength of 0.8 μm, a peak intensity of 1.3×10^{15} W cm^{-2}, and several values of j: (a) $j = j_{\min} = 66$ ($P^2/2m$ = photoelectron energy = 1.24 eV), (b) 67 (2.24 eV), and (c) 70 (5.24 eV). For spectrum (d), $j = 66$ and $I = 6.5 \times 10^{14}$ W cm^{-2}. In these calculations, $\vec{P}$ is assumed to be parallel to the pump polarization ($\hat{\varepsilon}$).

Obtaining the overall HG spectrum requires summing the contributions from all ATI processes and including the three transition matrix elements of Eq. (7). For a given harmonic order q, the matrix element can be expressed as [14]

$$
\begin{aligned}
T_q = 2\pi \int_0^{\pi} \sin\varphi \, d\varphi \sum_j e^{-ij\Delta\Theta}(z - j)\sqrt{2m^3\omega^5(j - z - E_b/\omega)}\left(-\frac{eg'}{2\Lambda}\right) \\
\times |\Phi_i(\vec{P}_j + z\vec{k} - j\vec{k})|^2 \left| \mathfrak{T}_{-j}\left(\zeta_j \cos\varphi, -\frac{z}{2}, 0\right)\right|^2 \\
\times \left\{-\zeta_j \cos\varphi \cdot \mathfrak{T}_{-q}\left(x_j \cos\varphi, y, -\frac{\pi}{4}\right) + 2ze^{-i\frac{\Delta\Theta+\pi}{2}}\mathfrak{T}_{-q+1}\left(x_j \cos\varphi, y, -\frac{\pi}{4}\right)\right. \\
\left. + 2ze^{i\frac{\Delta\Theta+\pi}{2}}\mathfrak{T}_{-q-1}\left(x_j \cos\varphi, y, -\frac{\pi}{4}\right)\right\},
\end{aligned}
\tag{16}
$$

where $\Phi(\vec{P} + z\vec{k} - j\vec{k})$ is the Fourier transform of the bound state wave function. Notice that the contributions from all photoelectrons, regardless of energy, add coherently. Furthermore, because $\mathfrak{T}(x, y, \phi)$ is an even or odd function of $\vec{P}$ (depending upon the order q), integrating over $\vec{P}$ results in the even harmonics vanishing.

Theoretical spectra, determined from Eq. (16) by calculating $\int d(\Delta\Theta)\,|T_q|^2$, are compared in the upper portion of Fig. 2 with a few of the results reported by Krause *et al.* [3] (KSK—Fig. 6) who determined the single atom response for He and H to a strong field by numerically integrating the time-dependent Schrödinger equation. Calculations are presented for two driving optical field wavelengths, 527 and 1053 nm, the peak intensity is taken to be 6×10^{14} W cm^{-2} in both cases, and the two pairs of spectra have arbitrarily been normalized at 40 eV. For the larger photon energy ($\hbar\omega = 2.35$ eV), the QED theory matches the predicted spectrum of Ref. [3] quite well. However, as $\hbar\omega$ is decreased, the QED spectral intensity does not fall as rapidly with increasing q as does the KSK spectrum and cutoff is ~25% lower in the QED calculation than that reported in Ref. [3]. As noted previously, cutoff in the QED framework is a consequence of a property of the generalized Bessel function that describes the coupling between the electron and the quantized laser field. Therefore, the physical origin for cutoff in the QED theory differs from the semiclassical view—the former can be interpreted as a decline in the Volkov-to-Volkov state transition probability.

The calculations by KSK [3] suggest that the onset of cutoff obeys the phenomenological relation $W_p = E_b + AU_p$, where $A \cong 3$. The lower portion of Fig. 2 illustrates the dependence of harmonic cutoff on $z = U_p/\hbar\omega$, calculated by QED theory for various values of j and assuming E_b to be 24.6 eV (He). For convenience, the expression $(3.17U_p + E_b)/\hbar\omega$ is also shown by the heavy solid line. Note that cutoff for large j values (~30, $\vec{P}^2/2m \approx 35$ eV) approaches the behavior proposed in Ref. [3] but including only the lowest photoelectron orders ($j_{\min}, j_{\min} + 1, \ldots$) yields a value for A closer to 2 (dashed line in Fig. 2).

Each ATI order correlates with a specific (discrete) photoelectron energy and the overall HG spectrum is determined by the degree of participation by each order. On

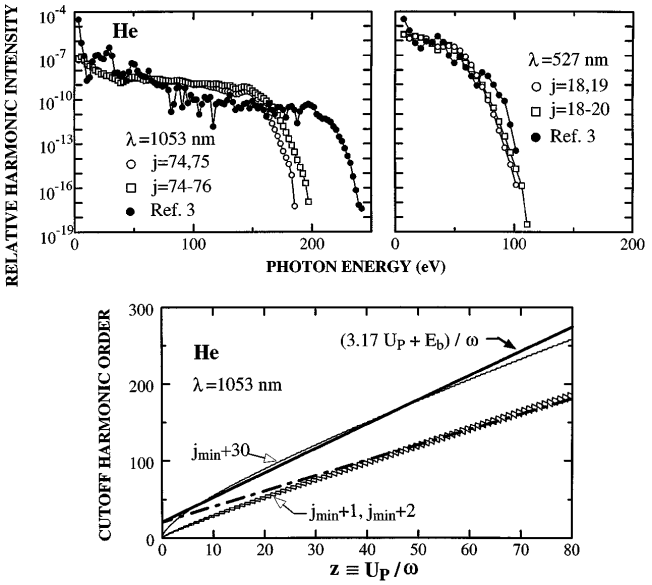


FIG. 2. Top panel: Comparison of the theory of Ref. [3] (●) for He with the QED theoretical results (○, □). Data for driving optical field wavelengths (λ) of 527 and 1053 nm and two sets of j values are shown. For $\lambda = 527$ nm, $j = 18$, 19 (○) and $j = 18-20$ (□). For the 1053 nm results, $j = 74$, 75 (○) and $j = 74-76$ (□). Bottom: Cutoff as a function of z. E_b is taken to be 24.6 eV (He), $\lambda = 1053$ nm ($\hbar\omega = 1.18$ eV), and results for several ATI photoelectron orders are shown and compared with the relation: $(3.17U_p + E_b)/\hbar\omega$. The dashed line represents the expression $(2U_p + E_b)/\hbar\omega$.

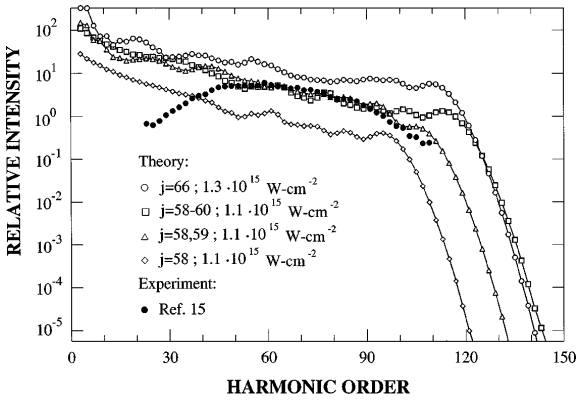


FIG. 3. Comparison of the theoretical harmonic generation spectrum with the experimental data of Macklin *et al.* (Ref. [15]) for Ne. The $I = 1.3 \times 10^{15}$ W cm^{-2} spectrum (○) assumes $j = j_{\min} = 66$, whereas three spectra are presented for an intensity of 1.1×10^{15} W cm^{-2}, which incorporate the following ATI orders: ◇: $j = 58$; △: $j = 58$, 59; □: $j = 58-60$. For all of the spectra, $\lambda = 0.8$ μm ($\hbar\omega = 1.55$ eV).

the basis of the computations to date, it appears that only the lowest order ATI electrons (i.e., $j_{\min} \rightarrow j_{\min} + 3$) contribute significantly to HG, as evidenced by comparisons with experimental spectra reported thus far. Preliminary calculations of the ellipticity of various harmonics confirm this conclusion. As an example, consider Fig. 3 which compares the results of Macklin *et al.* [15] in Ne with the predictions of QED theory. Three theoretical spectra are shown to illustrate the impact of pump intensity and the ATI photoelectron order on the result. For the intensity quoted in Ref. [15] (1.3×10^{15} W cm^{-2}), the match of the theoretical spectrum with the experimental data is poor for harmonic orders (q) above ≈ 87. If the laser intensity is taken to be $\sim 20\%$ smaller, however, then the data of Ref. [15] are reproduced well by including only the lowest two ATI orders. Although harmonics above 109 were not detected in Ref. [15], the roughly 1 order of magnitude decline in intensity between harmonics 87 and 109 that was observed experimentally is also a characteristic of the $j_{\min}, j_{\min+1} = 58, 59, I = 1.1 \times 10^{15}$ W cm^{-2} spectrum of Fig. 3. The key point to be made is that for intensities within $\pm 25\%$ of the published value [15], the data cannot be reconciled with experiment if j values $> j_{\min} + 2$ are incorporated into the calculations.

In summary, the theoretical framework presented here, derived from a QED perspective, offers a new interpretation of HG by (1) proposing that dressed continuum $\rightarrow$ continuum transitions in the form of transitions between quantum Volkov states represent the dominant contribution to the overall harmonic spectrum and (2) confirming the linkage between ATI and HG.

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